

Robust forecast evaluation under extreme shocks*

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Abstract

We address forecast evaluation in the presence of extreme, one-off shocks, such as the impact of Covid-19 on GDP. Conventional predictive ability tests may have no power under such brief but severe instabilities, potentially overlooking the superior accuracy of forecasts during normal periods. To remove this effect, we propose winsorizing forecast errors to neutralize the influence of extreme events. Using the US Survey of Professional Forecasters for real GDP growth, we illustrate how this approach uncovers superior performance that standard tests would otherwise fail to detect.

JEL codes: C22, C53 .

Keywords: Forecast evaluation, structural instability, winsorization, forecasting, SPF.

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1 Introduction

Instability of the forecasting performance with respect to a user-chosen metric, for example the mean squared forecast error, is an intrinsic characteristic of macroeconomic time series. As discussed in Rossi (2021), the sources may be different and include changes in the relations underpinning the economic model, or in the forecasting ability of economic predictors.

For output or for similar measures of economic activity, the empirical evidence in Chauvet and Potter (2013), and more recently in Siliverstovs and Wochner (2021) suggests that, while simple benchmarks are sufficient for the periods of expansion, recessions are more difficult to forecast and require more sophisticated procedures. For example, Ng and Wright (2013) show that forward-looking financial variables such as credit spreads improve GDP forecasts during the Great Financial Crisis, but they caution that this largely reflects the crisis' financial origins and may not generalize to earlier recessions, such as those of the early 1980s, which were driven by monetary policy tightening.

Compared with the recessions that preceded it, however, the economic instability triggered by the Covid-19 pandemic was different both because the scale of the shock vastly exceeded that of past downturns, and because the effect was very narrowly concentrated in time. It was also different, of course, because it did not originate from economic dynamics. Methods that until that point had proved successful in forecasting and nowcasting GDP or output did not cope well with the events in 2020 (see, e.g., Almuzara et al., 2023). Forecast breakdown is also documented outside the United States. In their study of Singapore, Chow and Choy (2023) show that professional forecasters' subjective uncertainty rose to record highs in the aftermath of the Covid-19 shock.

Major forms of instability can also affect the evaluation of forecasting performance. Rossi (2021) warned that instability can invalidate standard diagnostics tests, and Giacomini and Rossi (2010) showed that the fluctuation test is better suited to detect locally superior forecasting performance. However, Iacone, Rossini and Viselli (2026) showed that, in presence of extreme and very short-lived shocks, even tests like the fluctuation test are vulnerable and

could fail to detect a superior performance altogether, especially when the long run variance is estimated using the entire sample. Heuristically, this happens because sizeable shocks generate a noise that is so large that it covers the signal completely. In this case, pooling the sample is not advisable.

In this paper, we explore the possibility of removing this undesirable effect by “winsorizing” the observations of the loss differential for the period of extreme instability. Winsorising was originally introduced as a way to limit the influence of observations that are too extreme to be measured reliably, or that arise from heavy-tailed distributions (see, e.g., Tukey, 1962). Unlike outright removal of these observations, winsorizing preserves some information about both the magnitude and direction of the observation. This approach can reduce the potentially distorting effect of shocks on signal estimation—in our case, the loss differential. More specifically, we use it to reduce the variance of the estimate of expected loss in the Diebold and Mariano (1995) test, thereby retaining a significant test statistic when the expected loss differential is non-zero. The intuition behind the procedure is analogous to generalized least squares (GLS), which is useful to reduce the variance estimate and, in turn, can increase the power of the test on the parameter of interest. The duration of the Covid-19 pandemic, however, is too short to consistently estimate the variance of the shock period: capping the observations to a quantity that is comparable with the rest of the sample therefore is an appealing alternative. This approach is similar to the proposal in Lenza and Primiceri (2022), who suggested to weight down the influence of “Covid-19” observations in the estimation of the parameters (a VAR in their case) when forecasting GDP after the recovery. In a similar fashion, Schorfheide and Song (2024) recommend ignoring these observations altogether. In this sense, winsorising Covid-19 observations has already been used in spirit, through downweighting or exclusion, but our contribution is to formalise this idea for the loss differential within an explicit forecast-evaluation framework.

We briefly discuss winsorizing in section 2. In Section 3 we analyse the performance of the winsorised Diebold and Mariano (1995) test in a small Monte Carlo exercise, and in Section 4

we apply this test to evaluating forecasts of the real US GDP growth over a sample spanning the period of the Covid-19 pandemic. We conclude in Section 5.

2 The winsorized test statistic

We consider the classical Diebold and Mariano (1995) test for equal predictive ability, hereafter DM. In particular, we assume that we are interested in comparing two different forecasts of a series y_t on the basis of previously available information. Denoting these forecasts as $\hat{y}_t$ and $\tilde{y}_t$, and their respective errors as $\hat{e}_t = y_t - \hat{y}_t$ and $\tilde{e}_t = y_t - \tilde{y}_t$, for a loss function $L(\cdot)$, then the losses associated to these errors are $L(\hat{e}_t)$ and $L(\tilde{e}_t)$, respectively, and the loss differential is $d_t = L(\hat{e}_t) - L(\tilde{e}_t)$. Common loss functions that are popular in the literature are the squared loss function $L(e) = e^2$, the absolute loss $L(e) = |e|$; Giacomini and White (2006) also considered linex loss and other functional forms. As in Giacomini and White (2006) the loss function may be generic, as long as some regularity conditions are met. In our Monte Carlo exercise and in our empirical application, however, we will focus on the quadratic loss function, so the loss differential in those situations is

$$d_t = \hat{e}_t^2 - \tilde{e}_t^2.$$

In the DM framework, we test $H_0 : \{E(d_t) = 0 \text{ for all } t\}$ using the standardised sample mean $t_{DM} = \sqrt{T}\bar{d}/\hat{\sigma}$, where $\bar{d}$ is a sample average and $\hat{\sigma}^2$ is an estimate of the long-run variance σ^2 of d_t . When H_0 holds, assuming $\hat{\sigma}^2 \rightarrow_p \sigma^2$ and other regularity conditions as, for example, in Giacomini and White (2006), then $t_{DM} \rightarrow_d Z$ as $T \rightarrow \infty$, where Z is a standard normal random variable.

To appreciate the power of the test, we need to consider the alternative. The DM test can be interpreted as a test on the mean: denoting $\mu_t = E(d_t)$ and $u_t = d_t - \mu_t$ so that u_t is a zero mean process, the most natural alternative would be $H_A : \{\mu_t = \mu \neq 0\}$ for all t . Under regularity conditions (see West, 1996; Giacomini and White, 2006), this test performs

successfully in this situation. As it is common when evaluating tests, it is also possible to study the local power by considering a Pitman drift $\mu_t = \delta_1 T^{-1/2}$ as in Coroneo and Iacone (2020) (also see Giacomini and White, 2006, where alternatives decreasing in μ are considered).

Alternatively, it is possible that $H_A : \{\mu_t = \mu \neq 0\}$ only for some values of t . This is the situation considered in Giacomini and Rossi (2010). Denoting $I_t(\tau)$ as an indicator function taking value 1 if $t = \lfloor \tau T \rfloor$ and 0 otherwise, we can formalise this hypothesis as $H_A : \{\mu_t = \delta_2 I_t(\tau)\}$ with $\tau \in [\tau_1, \tau_2] \subset (0, 1)$ (and $\delta_2 \neq 0$). The power of the DM test wanes as the interval $[\tau_1, \tau_2]$ shrinks, whereas the fluctuation statistic, adapting the test to operate a DM test on just a fraction of the sample, effectively restores it.

The assumption that $\tau \in [\tau_1, \tau_2] \subset (0, 1)$ is key for the success of the fluctuation test, and in practice the span $\{\lfloor \tau_1 T \rfloor, \dots, \lfloor \tau_2 T \rfloor\}$ must be long enough to warrant any power. In fact, when $I_t(\tau) = 1$ for a very short period, possibly even just point in the sample, this instability may affect the DM test as a form of noise, and Iacone, Rossini and Viselli (2026) showed that when it is too large then it may even mask an otherwise significant signal.

This situation of interest can be described as

$$d_t = \delta_1 T^{-1/2} + \delta_2 T^\alpha I_t(\tau) + u_t, \quad (1)$$

where $\delta_2 T^\alpha$ is the scale of the instability relative to the sample size. Allowing for $\alpha > 0$, we include in this class also fairly large shocks, which realistically could describe situations like the 1973 oil shock or the 2008 Global Financial Crisis. Extreme shocks are represented by $\alpha \geq 1/2$ (and $\delta_2 \neq 0$). The DM null hypothesis corresponds to $\delta_1 = 0$, $\delta_2 = 0$, and $\delta_1 \neq 0$ characterizes the alternative $\{E(d_t) \neq 0 \text{ for all } t\}$.

Iacone, Rossini and Viselli (2026) examine cases with $\delta_2 \neq 0$, finding that shocks with scale $\alpha < 1/2$, do not affect the limit performance of the DM test. On the other hand, extreme shocks affect both the numerator and the denominator of the test statistic t_{DM} , driving the power towards zero for reasonable theoretical size. They document this loss of power both

through Monte Carlo simulations and in an empirical application to US GDP data.

2.1 Winsorized loss differential

To prevent this undesirable power loss in presence of extreme shocks, we propose to winsorize the effect of the shock $I_t(\tau)$ on the loss differential d_t . Denoting

$$d_{\inf} = \inf_{t \neq \lfloor \tau T \rfloor} \{d_t\}, \quad d_{\sup} = \sup_{t \neq \lfloor \tau T \rfloor} \{d_t\},$$

we set

$$\begin{aligned} d_{\lfloor \tau T \rfloor}^* &= d_{\inf}, & \text{if } d_{\lfloor \tau T \rfloor} < d_{\inf}, \\ d_{\lfloor \tau T \rfloor}^* &= d_{\sup}, & \text{if } d_{\lfloor \tau T \rfloor} > d_{\sup}, \\ d_{\lfloor \tau T \rfloor}^* &= d_{\lfloor \tau T \rfloor}, & \text{if } d_{\inf} \leq d_{\lfloor \tau T \rfloor} \leq d_{\sup}, \end{aligned}$$

and compute the test statistic using the series $\{d_1, \dots, d_{\lfloor \tau T \rfloor}^*, \dots, d_T\}$. We distinguish between the two statistics by denoting with t_{DM}^* the one computed using the winsorized series.

Importantly, here we assume that τ is known in advance. This seems realistic when dealing with the instability generated by the Covid-19, and this knowledge was exploited, for example, by Lenza and Primiceri (2025) and by Shorfeide and Song (2024) for the purpose of forecasting. Since we treat the location of the shock as known, it is possible that winsorization does not take place. More specifically, the probability that winsorization occurs when no shock actually takes place ($\delta_2 = 0$) is asymptotically negligible. When winsorization does not occur, the t_{DM} and t_{DM}^* statistics are exactly the same.

Heuristically, it follows that the asymptotic properties of the t_{DM}^* statistic are not affected when $\delta_2 = 0$. To appreciate the effect of the correction when $\delta_2 \neq 0$, notice that—following for example Phillips and Solo (1992), page 978, and Lemma E.4 of Ibragimov and Phillips (2008), along with the regularity conditions therein—we have $d_{\sup} = o_P(T^{1/2})$. Hence, asymptotically, the t_{DM}^* statistic is not affected by this shock. This means that winsorizing is a viable solution

to the problem. We formalize sufficient conditions and establish this result in the following subsection.

2.2 Asymptotic properties

Proceeding as in Iacone, Rossini and Viselli (2026), we study the limit properties of the winsorized DM statistic t_{DM}^* assuming that the long-run variance is estimated using the Bartlett kernel,

$$\hat{\sigma}^2 = \sum_{l=-M}^M c_l$$

where c_l is the l -th sample covariance of d_t^* and M is a user-chosen bandwidth such that $M/T \rightarrow 0$ as $T \rightarrow \infty$.

To establish the limit properties of the t_{DM}^* statistic we introduce the following assumption.

Assumption GW

(GW.1) u_t is mixing with ϕ of size $-r/(2r-2)$, $r \geq 2$; or α of size $-r/(r-2)$, $r > 2$;

(GW.2) $E(|u_t|^{2r}) < \infty$ for all t ;

(GW.3) $Var\left(\frac{\sqrt{T}}{T} \sum_{t=1}^T u_t\right) > 0$ for all T sufficiently large.

Assumption GW is based on Giacomini and White (2006) and is also used in Iacone, Rossini and Viselli (2025), so we refer to these papers for a discussion. The slightly stronger moment condition is a Lyapunov-type condition to establish a law of large numbers for $\sup(|u_t|^{2r})$ without assuming identical distribution. This is sufficient to account for the modification of the test statistic that is induced by winsorizing the observations.

To facilitate the characterization of the local power of the test, it is also convenient to assume the existence of $\sigma^2 = \lim_{T \rightarrow \infty} Var\left(\frac{\sqrt{T}}{T} \sum_{s=1}^T u_s\right)$.

Theorem 1. *Under Assumptions GW.1 – GW.3, and $r > 2$,*

$$t_{DM}^* \rightarrow_d Z + \frac{\delta_1}{\sigma}.$$

We refer to Appendix [A](#) for a complete proof.

We therefore consider testing $H_0 : \{E(d_t) = 0 \text{ for all } t \neq \lfloor \tau T \rfloor\}$ using the t_{DM}^* statistic, with critical values from the standard normal distribution, or from the modified limit distribution when fixed smoothing is applied, as in Coroneo and Iacone (2020).

In particular, we emphasise that the time point $t = \lfloor \tau T \rfloor$ is excluded from our null hypothesis, unlike in Diebold and Mariano (1995). This is because the winsorised test statistic is not informative about what happens at that specific point: winsorisation effectively removes the original observation. Inference on $E(d_{\lfloor \tau T \rfloor}) = 0$ can instead be conducted, for example, using the S statistic as in Iacone, Rossini and Viselli (2026).

In practice, the same argument justifies winsorization even when extreme instability affects more than a single point in time, provided that the number of such points in time is finite and small relative to the sample size, as it was the case during the Covid-related instability.

3 Monte Carlo study

We analyse the performance of the DM^* test in a small Monte Carlo experiment. For comparability, we adopt the same design as in Iacone, Rossini and Viselli (2026) and simulate

$$d_t = \kappa + \theta I_t(\tau; m) + u_t,$$

over a range of realistic parameter values, where u_t is a zero-mean stationary process and $I_t(\tau; m)$ is an indicator function equal to 1 when $t \in \{\lfloor \tau T \rfloor, \dots, \lfloor \tau T \rfloor + m - 1\}$, and m denotes the number of observations subject to instability. To avoid confusion with the theoretical situation described in Eq. (1), we distinguish the average, global difference in predictive accuracy $\mu_t = E(d_t)$ when $t \notin \{\lfloor \tau T \rfloor, \dots, \lfloor \tau T \rfloor + m - 1\}$ by denoting it as κ .

In line with the empirical application of interest, we fix $T = 100$ and $\lfloor \tau T \rfloor = 81$, and consider both $m = 1$ and $m = 3$. We estimate the long run variance using the Bartlett

Table 1: Empirical size: $\kappa = 0$ and $\theta = 0$.

ϕ	$m = 1$			$m = 3$		
	0.00	0.25	0.50	0.00	0.25	0.50
DM	0.0481	0.0557	0.0716	0.0481	0.0557	0.0716
DM*	0.0483	0.0557	0.0717	0.0484	0.0559	0.0718

Notes: The table reports rejection frequencies over 10,000 Monte Carlo replications. The sample size is $T = 100$, the bandwidth $M = 10$, and the fixed- b critical value is 2.2606.

weighted covariance estimate, with bandwidth $M = 10$, applying the rule $M = \lfloor T^{1/2} \rfloor$; we computed the critical value using the fixed- b limit distribution as in Coroneo and Iacone (2020). For the process u_t , we consider an autoregression:

$$u_t = \phi u_{t-1} + \varepsilon_t,$$

where ε_t is independently and normally distributed, with $E(\varepsilon_t) = 0$, and $E(\varepsilon_t) = \sigma^2$. We calibrate the parameters of the Monte Carlo simulation on an AR(1) fitted to the loss differential from our empirical application over the first 80 observations, and then scale κ and θ so that $\sigma^2 = 1$. In our baseline design, we set $\kappa = -0.8$, $\theta = -80$, and $\phi = 0.25$ with $m = 3$. We also consider alternative configurations to examine effects on empirical size and power.

For the size study, where $\kappa = 0$ and $\theta = 0$, we consider values for ϕ in $\{0, 0.25, 0.5\}$. For the power study, we set $\phi = 0$ to avoid interference from potential size distortion and further explore additional values of κ and θ . For every experiment we use two-sided alternatives. For each combination of parameters we consider 10,000 replications.

We report in Tables 1 and 2 the empirical size and power of the Monte Carlo simulation, where each entry represents the fraction of rejections of the null hypothesis. The labels DM and DM^* refer to the use of the t_{DM} and t_{DM}^* statistics, respectively.

The results for the case $\kappa = 0$ and $\theta = 0$ are summarized in Table 1: these confirm that in this situation the DM^* test has the same size performance as the DM , so the possibility of winsorizing does not affect size. Notice that in our exercise the fractions of rejections of the null hypothesis are always the same: this is not necessarily the case because, even when

Table 2: Empirical size and power: $\phi = 0$.

κ	θ	DM	DM*
Panel A: Covid-shock length $m = 1$			
0.00	-40	0.0000	0.0490
0.00	-80	0.0000	0.0490
-0.40	0	0.9592	0.9593
-0.80	0	1.0000	1.0000
-0.40	-40	0.2343	0.9723
-0.80	-40	0.9975	1.0000
-0.40	-80	0.0000	0.9723
-0.80	-80	0.1375	1.0000
0.40	-40	0.0000	0.9308
0.80	-40	0.0000	1.0000
Panel B: Covid-shock length $m = 3$			
0.00	-40	0.0000	0.0479
0.00	-80	0.0000	0.0479
-0.40	0	0.9592	0.9593
-0.80	0	1.0000	1.0000
-0.40	-40	0.0000	0.9753
-0.80	-40	0.0001	1.0000
-0.40	-80	0.0000	0.9753
-0.80	-80	0.0000	1.0000
0.40	-40	0.0000	0.7248
0.80	-40	0.0000	1.0000

Notes: The table reports rejection frequencies over 10,000 Monte Carlo replications. The sample size is $T = 100$, κ is the average, global loss differential, θ is the shock-specific loss differential (e.g., during the Covid-shock period), and m is the length of the shock period. The bandwidth $M = 10$, and the fixed- b critical value is 2.2606.

$\theta = 0$, it is possible that the maximum of d_t occurs exactly at $t = \lfloor \tau T \rfloor$ and winsorising still takes place. However, this would be a very improbable event and in our simulation, even in the cases when it occurred, it did not change the decision about rejecting the null.

The cases with $\kappa \neq 0$ and $\theta \neq 0$ are reported in Table 2. All these situations are deviations from the null hypothesis (and identify power) for the DM test; on the other hand, recalling that for the DM^* test the null hypothesis only refers to $t \neq \lfloor \tau T \rfloor$, then the specific situations characterised as $\kappa = 0$ and $\theta \neq 0$ should still be interpreted as empirical size for the DM^* test.

We also consider cases in which $\theta > 0$ and $\kappa < 0$: these configurations describe temporary forecast breakdowns in which relative accuracy is reversed during the instability period, so that the forecast that is globally less accurate performs better during the shock. As expected, in all these situations, we find that the power of the DM test is adversely affected by an extreme shock whereas winsorizing removes this effect, restoring power without compromising the size. Interestingly, the power loss of the DM test is more pronounced when κ and θ have opposite signs. As predicted by Theorem 1, this occurs because, in the $m = 1$ case, the relevant signal is $\frac{1}{T} \sum_t \mu_t = \kappa + \frac{1}{T} \theta$, which is attenuated when the two terms offset each other.

We complete the exercise with a simulation using a set of parameter values that are calibrated on the data employed in our empirical analysis, that is $\kappa = -0.8$, $\theta = -80$ and $\phi = 0.25$, $m = 3$. We find that the null hypothesis is only rejected in 0.12% of the cases when the t_{DM} statistic is applied, but it is always (100%) rejected when the t_{DM}^* is used.

4 Empirical application to US Real GDP growth

We illustrate how winsorization of the loss function can influence forecast evaluation by considering the nowcasts of quarterly real GDP growth from the Survey of Professional Forecasters (SPF) over the 2000-2024 period ($T = 100$ observations). The data consists of quarter-over-quarter growth rates, expressed in annualized percentage points; for a detailed description of the dataset, see Stark (2010).

We compare the predictive accuracy of the SPF forecasts against a benchmark of zero quarterly GDP growth. This is, of course, a very poor benchmark, as it even ignores the fact that real GDP growth is on average positive due to long-term economic expansion. The purpose of this exercise is to show that, in the presence of a very large shock, the DM test fails to establish superior forecasting performance, even relative to a clearly inferior benchmark, while winsorizing extreme events can restore the detection of the better forecast.

We plot the SPF and benchmark forecast errors in Figure 1, whereas summary statistics and the results of the DM test are displayed in Table 3. As we anticipated, disregarding the

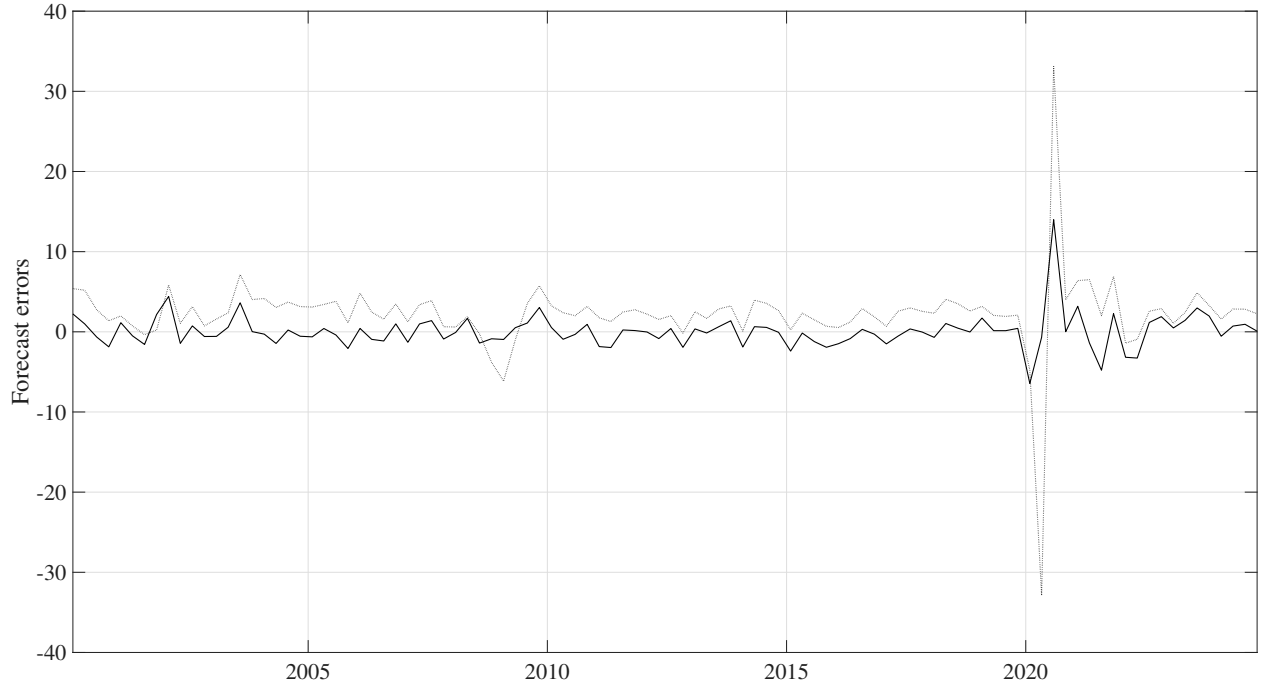


Figure 1: Errors associated to the quarterly SPF nowcasts (solid line) and the naive benchmark model (dotted line) over the period 2000:Q1 to 2024:Q4.

three observations for the Covid-19 period, the naive benchmark forecast error is on average positive (approximately 2.35) reflecting the growth rate of the real GDP over the period. In contrast, the average SPF forecast error is very close to zero. Their scores, in terms of the MSE, clearly favour the SPF forecasts during the ordinary period and this advantage becomes even more pronounced when the Covid-19 observations are included. Indeed, it is easy to verify that the ratios of the MSE and the root MSE between the naive benchmark and the SPF forecasts increase when the three COVID-19 observations are part of the sample.

Table 3: Forecast error summary and Diebold–Mariano test

Sample	Forecast	AFE	MSE	DM	DM*
Whole sample	SPF	0.0654	4.5118	1.458	6.147
	Naive	2.2313	31.1714		
Excluding pandemic	SPF	-0.0034	2.2010		
	Naive	2.3477	9.4551		

Notes: The table reports average forecast errors (AFE) and mean squared errors for the SPF Real GDP nowcasts and naive benchmark of zero growth. The pandemic period refers to 2020Q1–2020Q3. DM* denotes the winsorised Diebold–Mariano statistic. The 5% critical value is 2.261 (see Coroneo and Iacone, 2020).

Now, we discuss the results of the DM test. The test statistics were standardised using the Bartlett kernel estimate of the long run variance, with bandwidth set to $\lfloor T^{1/2} \rfloor$, and critical values for a two-sided test were obtained using fixed-smoothing asymptotics. The *DM* column reports the Diebold and Mariano statistic computed over the full sample, whereas the *DM** column shows the statistic after winsorizing the three observations for 2020:Q1 to 2020:Q3.

Consistently with the predictions from the theory and the findings of the Monte Carlo exercise, the canonical DM test fails to reject the null hypothesis, even though the loss function is on average non-zero at all points. Notably, this happens despite the SPF outperforming the naive benchmark even more strongly during the Covid-19 period, as indicated by the comparison of the square roots of the MSE in Table 3. The disappointing performance of the DM test is driven by its sensitivity to the extreme shocks that affected GDP during the Covid-19 period, a problem that is successfully addressed by winsorising those realisations of the loss function.

5 Conclusions

Extreme, short-lived shocks may adversely affect the power of standard tests of equal predictive ability, such as the Diebold and Mariano (1995) test, which may fail to detect systematic differences in forecasting performance. We show that winsorising the loss differential over a pre-specified instability window provides a simple correction to the DM statistic that restores power without altering the size properties of the original test. We support this result both theoretically and through Monte Carlo simulations.

We document the relevance of our approach in an empirical application to US real GDP growth and illustrate the practical usefulness of the procedure. The standard DM test fails to reject equal predictive ability even when SPF forecasts clearly outperform a simple benchmark outside the Covid-19 episode. After winsorising the pandemic observations, the evidence in favour of the SPF forecasts becomes much clearer. In applied work, the winsorised statistic

should therefore be reported alongside the standard DM statistic as a transparent robustness check when the sample contains rare, severe, and well-identified shocks.

Data sources

The series for the forecast errors of the growth rate of real GDP were downloaded from <https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/error-statistics> on 28/11/2025. A guide for the dataset and the paper of Stark (2010) were also available from the same page.

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Appendix A

Lemma A.1 Under the assumptions of Theorem 1,

$$\sup |u_t^2| = O_p(T^{1/p}).$$

for p such that $1 < p < r$.

Proof of Lemma A.1 We use an argument similar to (S.66) in Lemma C.2 of Iacone, Leybourne and Taylor (2019). For p such that $1 < p < r$, notice that $E(|u_t^2|^p) = E(|u_t|^{2p}) < \infty$, therefore

$$\sup |u_t^2|^p \leq \sum |u_t^2|^p = O_p(T)$$

using a WLLN for mixing sequences as in White (2001), Corollary 3.48. Next, notice that $\sup |u_t^2|^p = (\sup |u_t^2|)^p$ to establish the result.

Proof of Theorem 1 Denoting with $t_{DM^{**}}$ the test statistic computed from the series $\{u_1 + \delta_1 T^{-1/2}, \dots, u_{\lfloor \tau T \rfloor} + \delta_1 T^{-1/2}, \dots, u_T + \delta_1 T^{-1/2}\}$, Giacomini and White (2006) showed that $t_{DM^{**}} \rightarrow_d Z + \frac{\delta_1}{\sigma}$. If the winsorisation takes place, this corresponds to using the series $\{u_1 + \delta_1 T^{-1/2}, \dots, (u^* - u_{\lfloor \tau T \rfloor}) + u_{\lfloor \tau T \rfloor} + \delta_1 T^{-1/2}, \dots, u_T + \delta_1 T^{-1/2}\}$ where u^* is $\sup_t \{u_t\}$ or $\inf_t \{u_t\}$.

Denoting

$$\begin{aligned} \nu_t &= (u^* - u_{\lfloor \tau T \rfloor}) I_t(\tau), \\ \bar{\nu} &= \frac{1}{T} \sum_{t=1}^T \nu_t, \\ c_l(\nu\nu) &= \frac{1}{T} \sum_{t=l+1}^T (\nu_s - \bar{\nu})(\nu_{t-l} - \bar{\nu}), \end{aligned}$$

proceeding as in Iacone, Rossini and Viselli (2026), we need to show that $\sqrt{T}\bar{\nu} = o_p(1)$ to account for the numerator in the t_{DM^*} statistic, and $\sum_{l=-M}^M \frac{M-l}{M} c_l(\nu\nu) = o_p(1)$ to account for the denominator.

For the numerator, the condition is established using $u^* = o_p(T^{1/2})$, which follows from $\sup |u_t| = O_p(T^{1/(2p)})$.

For the denominator, notice that

$$\begin{aligned}\bar{\nu} &= \frac{1}{T}(u^* - u_{\lfloor \tau T \rfloor}), \\ \bar{\nu}^2 &= \frac{1}{T^2}((u^*)^2 + (u_{\lfloor \tau T \rfloor})^2 - 2u^* u_{\lfloor \tau T \rfloor}) = O_p\left(\frac{1}{T^2}T^{1/p}\right),\end{aligned}$$

so

$$\begin{aligned}c_0(\nu\nu) &= \frac{1}{T}(u^*)^2 + O_p\left(\frac{1}{T^2}T^{1/p}\right), \\ c_l(\nu\nu) &= O_p\left(\frac{1}{T^2}T^{1/p}\right) \text{ for } l \neq 0,\end{aligned}$$

and

$$\sum_{l=-M}^M \frac{M-l}{M} c_l(\nu\nu) = O_p\left(\frac{T^{1/p}}{T} + \frac{M}{T} \frac{T^{1/p}}{T}\right) = o_p(1).$$